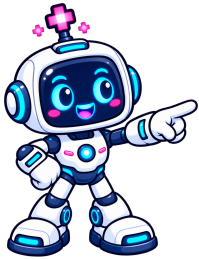




# ASSEMBLE YOUR CONTROLLER

MODEL-ACCURATE WORKBENCH ASSEMBLY

1 / 8



Pick the enclosure you printed, gather its parts, then follow that path without page hopping.



**UNO Q CASE**

**PAGE 3**

Compact base and lid. The Arduino hub remains outside.



**DOCK V1**

**PAGES 4-5**

Original low-profile base with an open rear hub bay.



**DOCK V2.1**

**PAGES 6-7**

Revised closed enclosure with room for real cables and plugs.



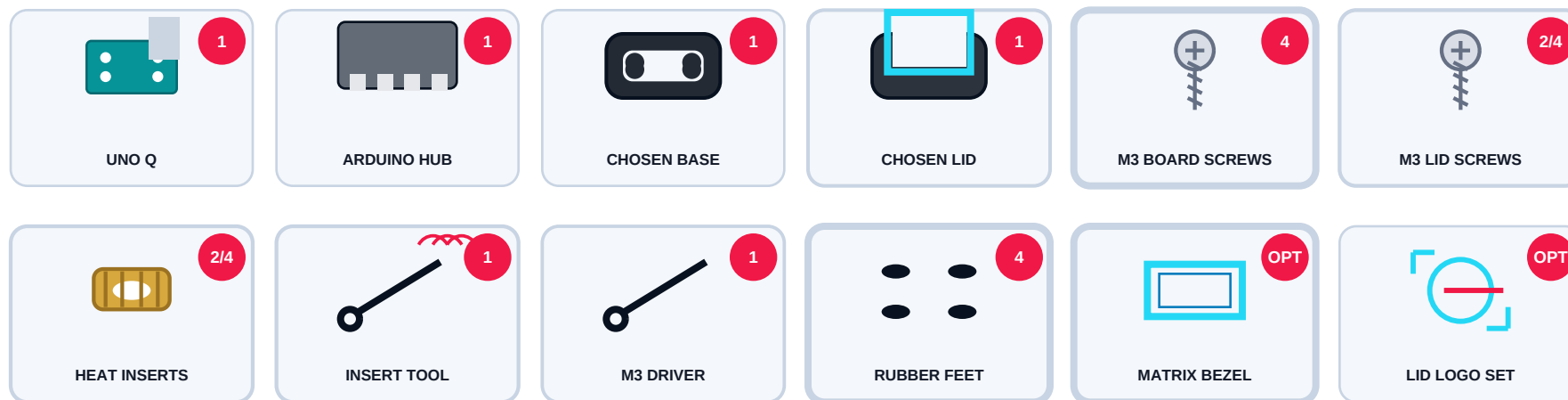
# PARTS, TOOLS + POWER OFF

MODEL-ACCURATE WORKBENCH ASSEMBLY

2 / 8

Print the fit coupons first. Assemble only with power disconnected.

For lid screws and heat inserts: 2 for the UNO Q Case, 4 for either Dock.



## HOT INSERT TOOL

Seat inserts square and flush. Stop before softened PLA deforms.



## NO FORCING

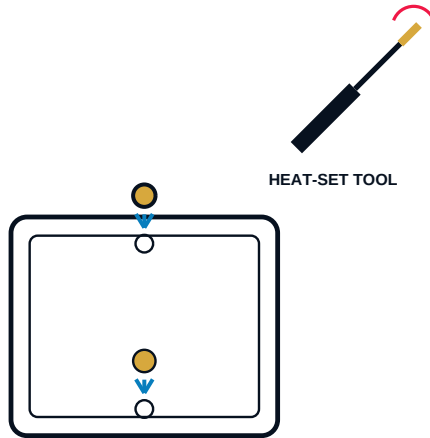
A plug, hub, board, or lid that resists needs a fit check - not more pressure.



## OPTIONAL PARTS

Foam, TPU, Ethernet, adhesive, bezel, and logo pieces are not structural.

x2



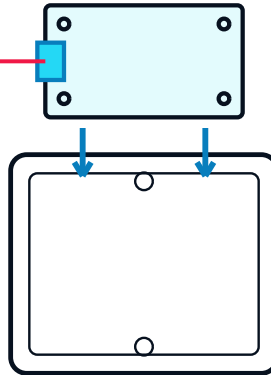
HEAT-SET TOOL

1

## Seat the two inserts

Use the two centreline bosses; keep each insert square and flush.

USB-C

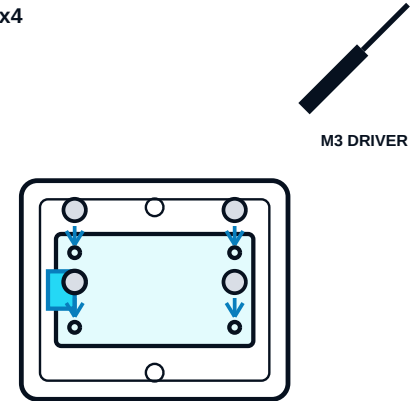


2

## Place the UNO Q

USB-C faces the broad case opening; the board lies flat.

M3 x 8 / x4

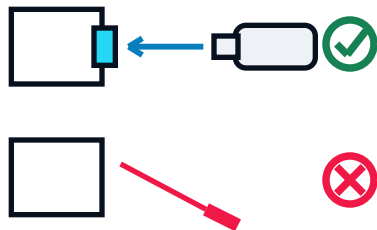


M3 DRIVER

3

## Fasten the board

Use four M3 x 8 mm screws. Snug, never bend the board.



4

## Check USB-C clearance

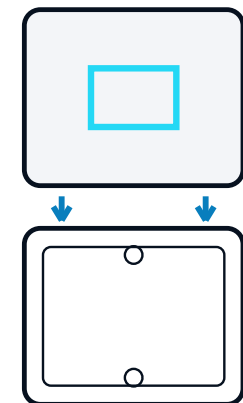
Connect the external hub; the plug must not push sideways.



5

## Choose and dry-fit branding

Use one matching compact-case logo set. Dry-fit backing, cyan, and red pieces before glue.

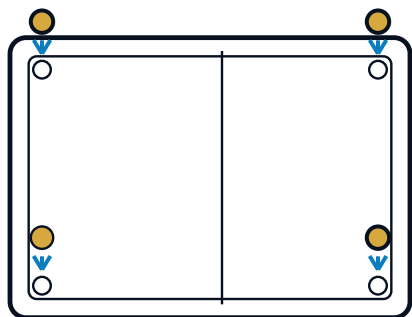


6

## Close and add feet

Fasten the lid, add four rubber feet, then inspect on page 8.

x4

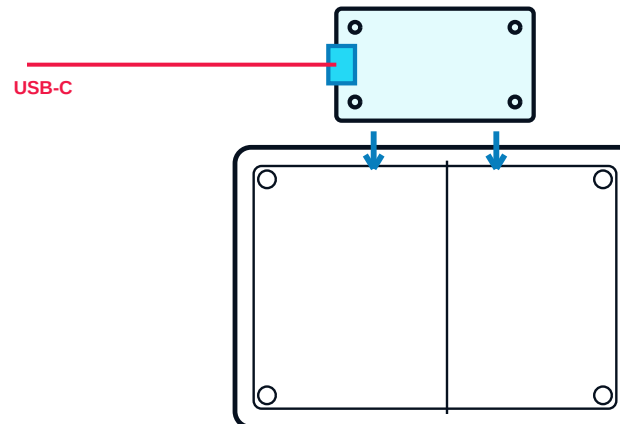


HEAT-SET TOOL

1

## Seat the four inserts

Keep each M3 insert square and flush with the boss.

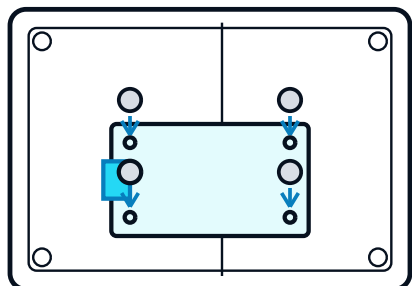


2

## Place the UNO Q

USB-C faces its broad opening and every mounting hole aligns.

M3 x 8 / x4

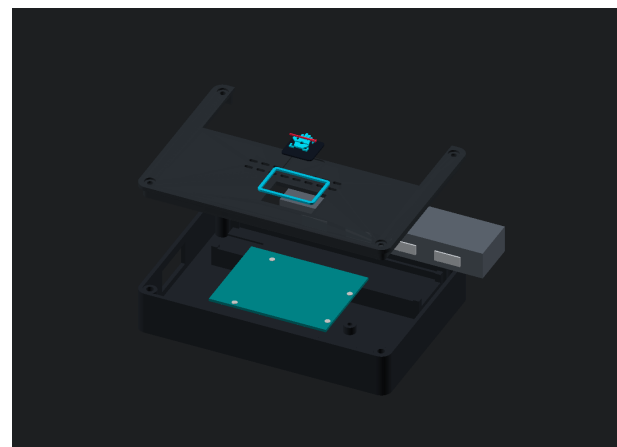


M3 DRIVER

3

## Fasten the board

Use four M3 x 8 mm screws without twisting the board.



4

## Add optional cradle padding

Use only a thin foam or TPU strip beneath the hub.



# CONTROLLER DOCK V1 - STEPS 5-8

MODEL-ACCURATE WORKBENCH ASSEMBLY

5 / 8

ROUTE THROUGH CHANNEL



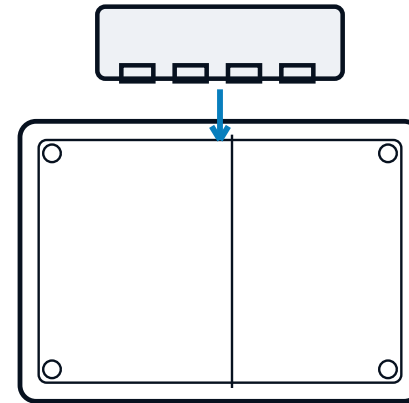
NO TWIST OR SHARP BEND



5

## Connect the captive cable

Route it through the channel and into the UNO Q without a twist.



PORTS OUT

6

## Seat the hub

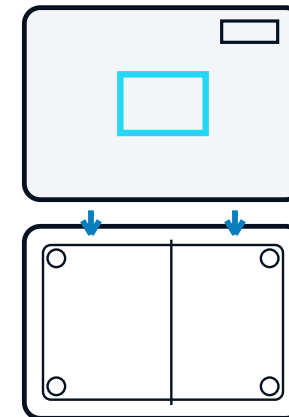
Lower it into the open rear bay; do not trap the captive lead.



7

## Connect the fixed cable set

Camera to USB-A 3.0, power to USB-C PD, Ethernet if used.

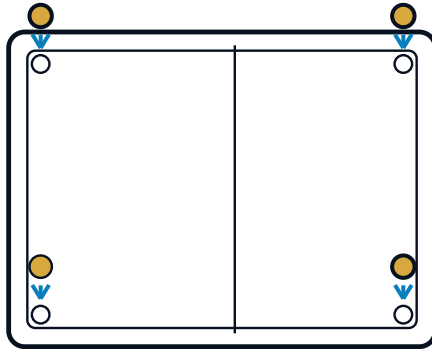


8

## Close and finish

Fit the lid, Matrix bezel, one matching logo set, and four rubber feet.

x4

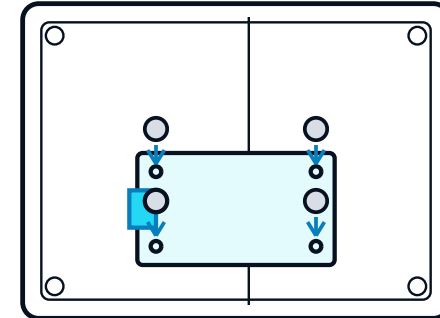


1

## Seat the four inserts

Keep each M3 insert square and flush with the corner boss.

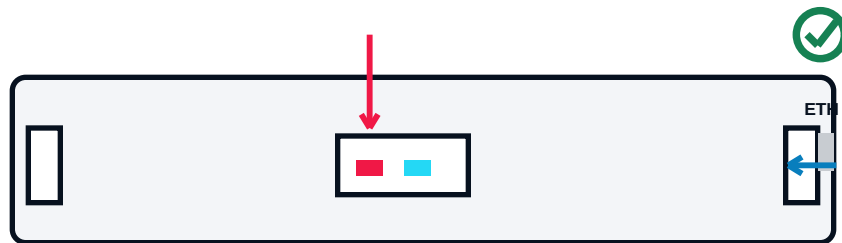
M3 x 8 / x4



2

## Mount the UNO Q

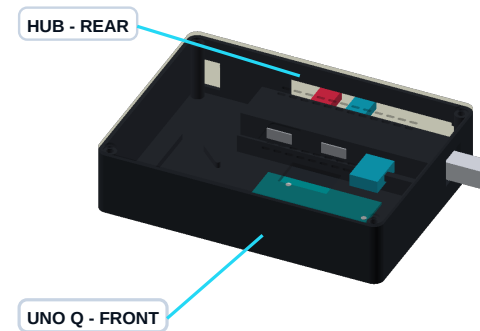
USB-C faces its broad opening; fasten with four M3 x 8 mm screws.



3

## Orient the hidden hub

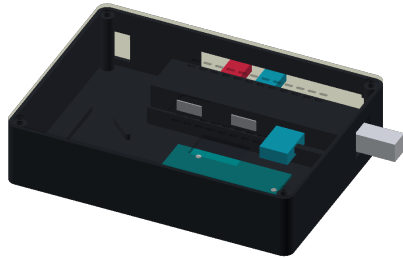
USB-C bank faces rear; Ethernet faces the right-side opening.



4

## Connect the captive cable

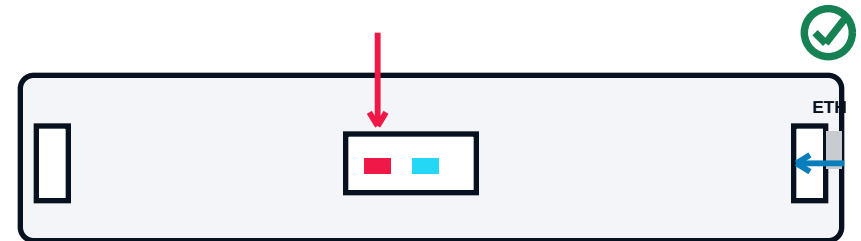
The OpenSCAD layout puts the hub behind the UNO Q. Route its lead without socket strain.



5

## Choose the camera route

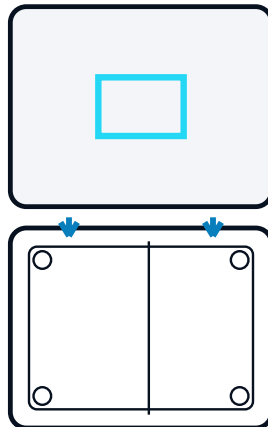
Plug USB-A inside and route its cable out, or use rear USB-C data.



6

## Keep power and Ethernet clear

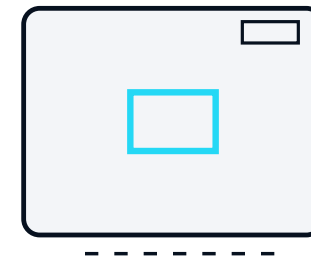
Power uses rear USB-C PD; Ethernet reaches the right-side opening.



7

## Dry-fit the lid

It must close freely without touching hub, plugs, cables, or board.



8

## Close and finish

Fasten the lid, fit the Matrix bezel and matching logo set, then add four feet.

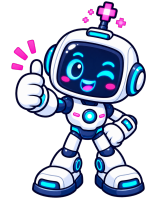


# INSPECT, POWER ON + PROVE IT

MODEL-ACCURATE WORKBENCH ASSEMBLY

8 / 8

The lid closes by itself. Screws hold it down - they do not force it down.



## BOARD FLAT

All four standoffs support the UNO Q; no screw twists it.



## NOTHING LOOSE

Remove spare screws, wire strands, and printed debris.



## PORTS CLEAR

Check Matrix window, vents, USB-C, cable routes, and Ethernet.



## NO SIDE LOAD

The hub cable must not push sideways on the UNO Q socket.



## LID CLOSURES FREELY

Stop if the lid touches a cable, plug, hub, or board.

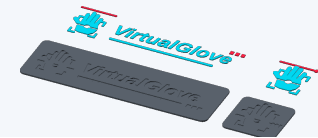


## 30-MINUTE TEST

Run the camera. Stop for soft PLA, excess heat, or USB drops.



CHOOSE ONE LID LOGO



BACKING + CYAN + RED; DRY-FIT BEFORE GLUE